

Scope

Q3.

a. GLOBAL SCOPE

```

└─ f
└─ main
    
```

main SCOPE

```

└─ argc, argv
└─ x = 2
    
```

f SCOPE

```

└─ y = 2
└─ x = y + 1 = 3
└─ for-loop SCOPE
    └─ i = 0, 1, 2
        └─ loop-body SCOPE
            └─ temp = i * 2 + x = 3, 5, 7
    
```

b. GLOBAL SCOPE

```

└─ main
    
```

main SCOPE

```

└─ a = 5
└─ ref_to_A_in_A → a
└─ inner block SCOPE
    └─ a = 10
    
```

c. GLOBAL SCOPE

```

└─ main
    
```

main SCOPE

```

└─ p → heap cell [42]
└─ inner block SCOPE
    └─ q → same heap cell [42]
    
```

Heap (outside SCOPE)

```

└─ [42] new/DELETE
    
```